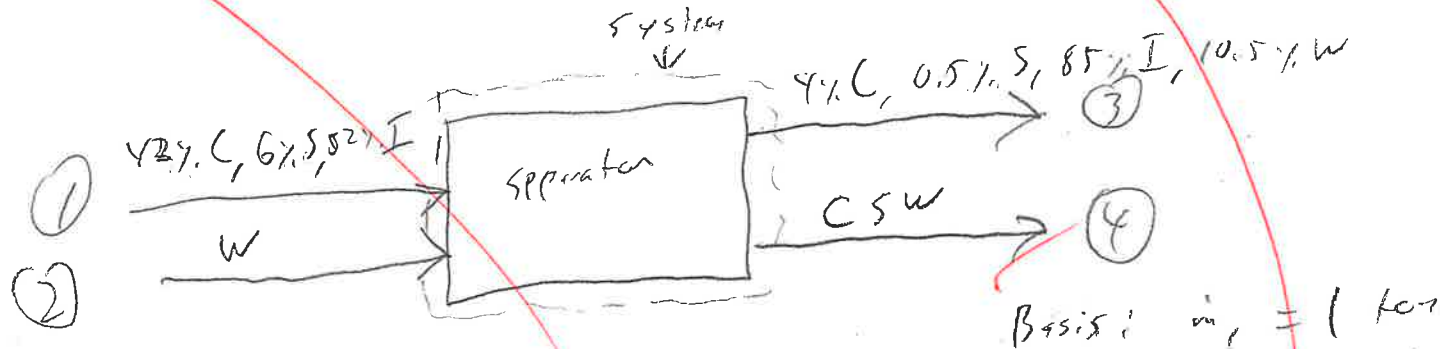


To Receive Full Credit, You Must Show All Your Work

The LeBlanc process produces "black ash" which is a solid consisting of 42 wt% sodium carbonate (C), 6 wt% water-soluble material (S), and 52 wt% water-insoluble material (I). The black ash is treated with liquid water (W) to extract the sodium carbonate. The result of the process produces a solid residue containing: 4 wt% C, 0.5 wt% S, 85 wt% I, and 10.5 wt% W. Also, a liquid product containing C, S, and W is produced. Use a basis of $m_1 = 1$ ton of black ash. Label the mass of liquid water added as m_2 , the mass of the solid residue produced as m_3 , and the mass of the liquid product m_4 . Give all your answers to the tenth of a lb_m .

- (a) Draw and label a block flow diagram. Define a system.
 (b) Calculate the mass of the solid residue, m_3 , the mass of the sodium carbonate extracted to the liquid product, $m_{4,C}$, and the mass of S extracted to the liquid product, $m_{4,S}$, in lb_m .
 (c) If the extraction process requires that the liquid product to be 75 wt% water, what is the mass in lb_m of the required liquid water added to the black ash, that is, m_2 , and the mass of the liquid product, m_4 ?



$$C: 0.42 \dot{m}_1 = 0.04 \dot{m}_3 + W_C \dot{m}_4$$

$$S: 0.06 \dot{m}_1 = 0.005 \dot{m}_3 + W_S \dot{m}_4$$

$$I: 0.52 \dot{m}_1 = 0.85 \dot{m}_3$$

$$W: \dot{m}_2 = 0.105 \dot{m}_3 + W_W \dot{m}_4$$

$$\dot{m}_3 = \frac{0.52}{0.85} \dot{m}_1 = 0.612 \text{ tons} = 1223.5 \text{ lb}_m$$

$$S: 0.06 \text{ tons} = 0.00306 \text{ tons} + W_S \dot{m}_4$$

$$C: 0.42 \text{ tons} = 0.02447 \text{ tons} + W_C \dot{m}_4$$

$$W: \dot{m}_2 = 0.064235 \text{ tons} + W_W \dot{m}_4$$

$$\dot{m}_{4,S} = 0.06 \text{ tons} - 0.5694 \text{ tons}$$

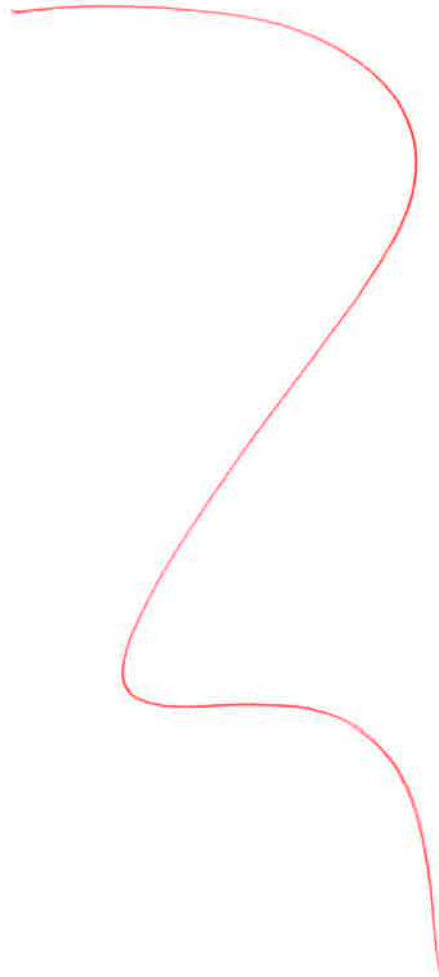
$$\dot{m}_{4,C} = 0.42 \text{ tons} - 0.0306 \text{ tons}$$

DoFA

$$\frac{11}{+0}$$

$$1 + \cancel{5} + 4$$

1 degree freedom



$$c) \quad 0.75 = \frac{\dot{m}_2 - 0.105 \dot{m}_3}{W_w} = \dot{m}_{yw} \quad ?$$

$$\dot{m}_{ys} = 0.15 \dot{m}_4 - \dot{m}_{4c}$$

$$\dot{m}_{4c} = 0.15 \dot{m}_4 - \dot{m}_{ys} \quad ?$$

$$W_c \dot{m}_4 = \underline{0.42 \dot{m}_1 - 0.04 \dot{m}_3}$$

$$W_s \dot{m}_4 =$$

